Event-Triggered Topology Discovery and Hold-Time-Aware Forwarding for Mobile USV–AUV Acoustic Networks

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***Abstract*—Mobile unmanned-surface-vehicle (USV) and autonomous-underwater-vehicle (AUV) teams must keep a usable acoustic topology while remaining inside a two-kilobit-per-second control budget. Recent underwater routing studies, including Q-learning designs such as LQARP, typically treat neighbor tables and link-quality estimates as already accurate, or they score connectivity with a geometric range disc. This paper treats topology discovery as the first object of study and then forwards on the map that discovery actually produces. We implement a sonar-equation link model—Thorp absorption, practical spreading, Wenz noise, and a BPSK packet-success gate—on Aqua-Sim-NG in NS-3, and we let HELLO beacons contend on unslotted ALOHA. An event-triggered send-on-delta policy with an SNR-aware timeout is compared with periodic beaconing across thirty randomized seeds. Event-triggered discovery at $T_{\max} = 25\,\mathrm{s}$ reduces control traffic from $927\,\mathrm{bit/s}$ to $212\,\mathrm{bit/s}$ (about 11% of a $2\,\mathrm{kbps}$ budget) while keeping precision at 0.994; the cost is recall 0.841 versus 0.966. On a relay-stable inner/outer field, the proposed ETHA protocol (event-triggered HELLO, hold-time filter, and depth-primary next-hop scoring) delivers packet delivery ratio 0.755 ± 0.031 at $2.82\,\mathrm{kJ}$, against 0.641 ± 0.027 at $4.61\,\mathrm{kJ}$ for periodic SNR-greedy forwarding ($n = 30$, 12 nodes, 95% Student-t intervals). The same event map with SNR-greedy scoring reaches only 0.651, so geography—not HELLO density—is the first-order PDR lever once the table is good enough. An oracle with perfect geometry reaches 0.817, locating the residual in missing relays rather than in hop scoring. The results argue for spending saved HELLO bits on a usable map before importing reinforcement-learning routing.**

***Index Terms*—Underwater acoustic networks, AUV, USV, topology discovery, event-triggered control, link-quality estimation, hold time, ALOHA, Aqua-Sim-NG, NS-3.**

Note to Practitioners: Ocean-IoT deployments that mix a USV sink with a small AUV team cannot copy Wi-Fi HELLO intervals. At $1\,\mathrm{kbps}$ BPSK, a 50-byte beacon every $5\,\mathrm{s}$ already consumes half of a $2\,\mathrm{kbps}$ control budget once twelve nodes share unslotted ALOHA. This paper gives operators a measured trade. For topology awareness, event-triggered beacons at $T_{\max} = 25\,\mathrm{s}$ hold that budget to about 11%, at the cost of slower link-up ($15\,\mathrm{s}$) and a sparser neighbor table. For data delivery, spend the saved bits on a shorter keep-alive ($T_{\max} = 10\,\mathrm{s}$) and a depth-primary next hop toward the surface sink: ETHA then matches a DBR-like baseline at $\mathrm{PDR} = 0.755$ while cutting transmit energy by about 39% relative to periodic SNR-greedy HELLO. Hold-time filtering stops dying relays; on a relay-stable field it rarely binds, and that coincidence is reported rather than advertised as a PDR miracle. The implementation is Aqua-Sim-NG/NS-3 and does not require sea-trial CSI or a Bellhop field.

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I. Introduction

UNDERWATER acoustic links are slow, half-duplex, and range-limited by absorption that grows with frequency [3]–[5]. A surface craft acting as a sink for a team of AUVs therefore cannot afford the HELLO rates that terrestrial mesh networks take for granted: a $2\,\mathrm{kbps}$-class control budget is already a large fraction of a $1\,\mathrm{kbps}$ BPSK pipe [1], [2], [17]. At the same time the geometry is not static. Meandering currents and AUV thrust move nodes across the decodable-range contour on a timescale of tens of seconds [8], so a neighbor table that is not refreshed becomes both incomplete and stale.

Most underwater routing literature evaluates packet delivery ratio (PDR) on a protocol that already assumes a neighborhood. Vector-based forwarding (VBF) and depth-based routing (DBR) need positions or depths [13], [14]. QELAR and the more recent LQARP protocol learn next-hop Q-values from rewards that mix energy, depth, and a link-quality term [16], [17]. Those designs are the right *second* chapter of a mobile USV–AUV network. They are the wrong first chapter if the input map is a geometric disc, an offline Bellhop table for moored nodes, or a HELLO process that has never been scored against collisions and missed detections [19]–[21].

This paper writes that first chapter. The question is mechanistic: under a realistic acoustic budget, how should a mobile node spend HELLO bits in order to know who is currently reachable, how good the link is, and how long it will remain usable—and does that knowledge change the next hop toward a USV sink? We answer it with a simulation-only Aqua-Sim-NG/NS-3 campaign [18], [30], [31] that is aligned with the acoustic stack of LQARP (source level, carrier, Wenz shipping and wind, SNR threshold $8.68\,\mathrm{dB}$, 400-bit packets) without copying its Q-learning router and without claiming a Bellhop transmission-loss table we do not run.

The contributions are as follows.

1) *A closed-form system model that matches the simulator.* Section II records the architecture, the Urick–Thorp–Wenz channel, the multipath stand-in, and the Q-learning construction that LQARP uses. Section IV then writes the graph, BPSK gate, Caruso kinematics, unslotted ALOHA, and transmit energy as the equations Aqua-Sim-NG actually integrates. A constant multipath margin

stands in for Bellhop; it is treated as a sensitivity parameter, not as a ray trace.

2) *Event-triggered, SNR-gated discovery under ALOHA.* HELLO beacons contend in Aqua-Sim unslotted ALOHA. Decode uses the sonar equation rather than a range disc. A send-on-delta policy in the sense of event-triggered control [9], [10] is compared with periodic beaconing on thirty randomized seeds with 95% Student-t confidence intervals.
3) *ETHA: hold-time-aware forwarding on the discovered map.* Each neighbor record carries an exponentially weighted moving average (EWMA) of SNR [28] and a remaining useful time obtained from radial opening speed and the SNR-threshold range $R^\star$. The proposed next hop applies a hold-time filter and then a depth-primary score toward the surface sink, using only neighbors the node has actually heard.
4) *A measured PDR and energy result, not a geometric claim.* On twelve nodes with $T_{\max} = 10\,\mathrm{s}$, ETHA reaches $\mathrm{PDR} = 0.755 \pm 0.031$ at $2.82\,\mathrm{kJ}$ and $480\,\mathrm{bit/s}$ of HELLO, versus 0.641 ± 0.027 at $4.61\,\mathrm{kJ}$ and $927\,\mathrm{bit/s}$ for periodic SNR-greedy forwarding. An oracle bound of 0.817 shows that the remaining loss is missing relays. Q-learning the next hop would be fitting the wrong residual.

The rest of the paper is organized as LQARP is organized. Section II defines the network architecture, the underwater acoustic propagation model, multipath as we treat it, ambient noise, and the Q-learning technique. Section III locates prior protocols against that background. Section IV instantiates the channel in the simulator and states mobility, MAC, energy, and the design problem. Section V presents event-triggered discovery and hold-time-aware forwarding (ETHA). Section VI reports the Aqua-Sim campaign. Section VII concludes.

II. Preliminaries

This section is definitional. It records the architecture of the cell, the closed-form acoustics that every later equation uses, the sense in which multipath is (and is not) present, the Wenz noise field, and the Q-learning loop that QELAR and LQARP train [16], [17]. No Q-table is estimated here, and no Bellhop eigenray is computed. The empirical claims of the paper begin in Section VI; they are not smuggled into these definitions.

A. Network Architecture

A mobile ocean-IoT cell consists of one unmanned surface vehicle (USV) and a small team of autonomous underwater vehicles (AUVs) that share a single half-duplex acoustic carrier [1], [2]. Index 0 denotes the USV. It rides near the surface and is the unique sink of every data packet. Indices $1, \ldots, N$ denote AUVs. Nodes cannot transmit and receive at once, and they cannot occupy a second frequency: control HELLO beacons and unicast DATA therefore contend for the same $1\,\mathrm{kbps}$ BPSK pipe.

Fig. 1 is the schematic. The USV is a surface gateway with a downward-looking hydrophone. AUVs occupy a water column whose depths, in this work, are drawn once in $[40, 160]\,\mathrm{m}$ and then held; the USV sits at $z = 8\,\mathrm{m}$. Inner AUVs lie inside the SNR-threshold range of the USV and can act as relays. Outer AUVs lie outside that range and require at least one underwater hop. Horizontal motion is the superposition of a meandering current [8] and a bounded AUV thrust; the figure marks it only as a drift. HELLO is a one-hop broadcast. DATA is a unicast toward the USV. There is no satellite backhaul in the model, no multi-USV backbone, and no acoustic modem other than the Aqua-Sim-NG device used in Section VI.

The architecture is deliberately small. Twelve nodes is already a crowded ALOHA neighborhood at $1\,\mathrm{kbps}$ [19]. The schematic is not a fleet exercise. It makes the objects of discovery and forwarding concrete before any protocol is written: who can currently be heard, who can currently relay toward the surface, and which of those facts a node is entitled to know.

B. Underwater Acoustic Propagation Model

Underwater links are slow, absorptive, and range-limited [3]–[5]. The workhorse abstraction remains Urick's passive sonar equation [5]

$$\gamma = \mathrm{SL} - \mathrm{TL} - \mathrm{NL} + \mathrm{DI} \qquad (\mathrm{dB}), \tag{1}$$

where SL is the source level, TL the transmission loss, NL the noise level in the receiver band, and DI the directivity index. We take $\mathrm{SL} = 185\,\mathrm{dB}$ re $1\,\mu\mathrm{Pa}$ at $1\,\mathrm{m}$ and $\mathrm{DI} = 0\,\mathrm{dB}$, as in LQARP Table II [17]. The quantity γ is an energy SNR. It is not a realized channel impulse response.

Let d be the Euclidean range in metres and f the carrier in kHz. Thorp's absorption, in dB/km, is [3], [6]

$$\alpha(f) = \frac{0.11 f^2}{1 + f^2} + \frac{44 f^2}{4100 + f^2} + 2.75 \times 10^{-4} f^2 + 0.003. \tag{2}$$

Practical spreading with exponent $k = 1.5$ interpolates between spherical ($k = 2$) and cylindrical ($k = 1$) spreading [4]. Together with a constant multipath margin M defined in Section II-C,

$$\mathrm{TL}(d) = 10k \log_{10} d + \alpha(f)\,\frac{d}{1000} + M, \qquad d \geq 1\,\mathrm{m}. \tag{3}$$

Sound speed is held at $c = 1500\,\mathrm{m/s}$. There is no sound-speed profile and no refraction. Propagation delay is d/c.

Substituting (3) into (1) gives the range-dependent SNR $\gamma(d)$. A packet is declared decodable when $\gamma(d) \geq \gamma_{\mathrm{th}}$ with $\gamma_{\mathrm{th}} = 8.68\,\mathrm{dB}$ [17]. The unique range

$$R^\star = \arg_d \{\gamma(d) = \gamma_{\mathrm{th}}\} \tag{4}$$

is then well defined for each (f, k, M, NL). At the Table II carrier $f_c = 24\,\mathrm{kHz}$ and the operating margin $M = 30\,\mathrm{dB}$ used later, $R^\star = 4946\,\mathrm{m}$. That number is a root of (1)–(3). It is not a sea-trial measurement.

C. Multipath Effects

A shallow waveguide does not deliver one arrival. A transmitted pulse reaches the hydrophone as a sum of eigenrays with different delays, amplitudes, and Doppler shifts [22], [23]. Two consequences matter for a 400-bit packet at $1\,\mathrm{kbps}$.

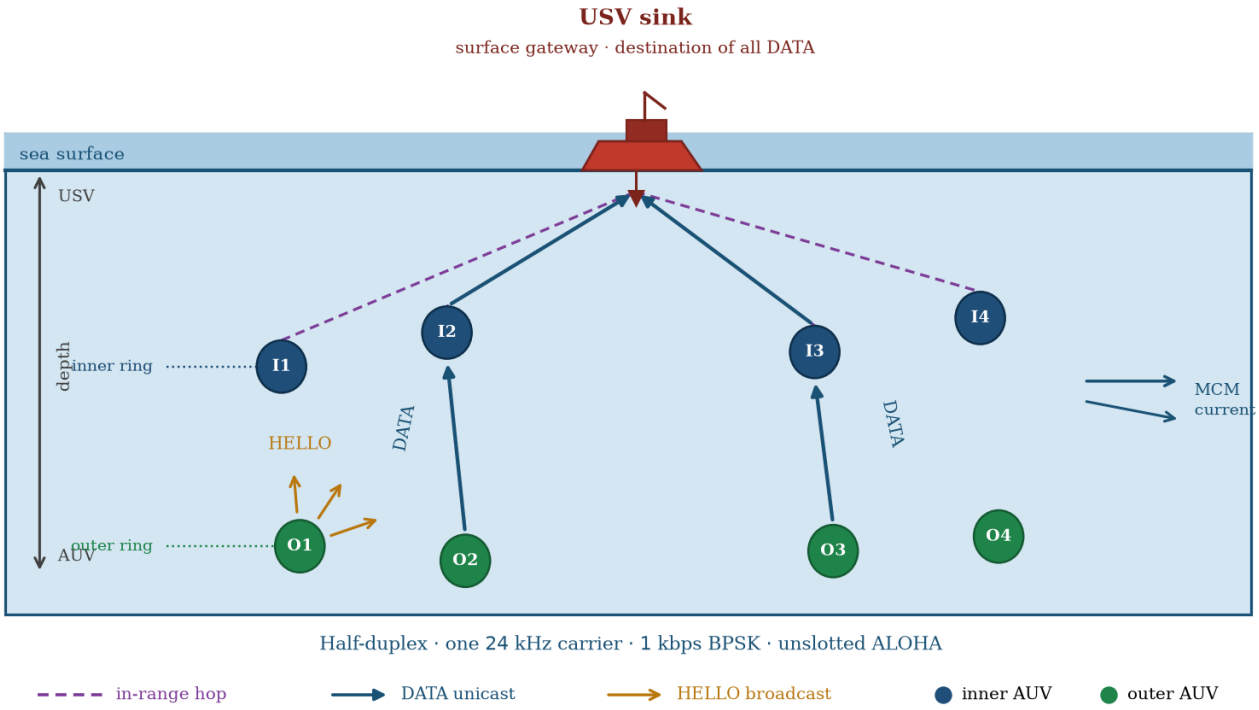


Fig. 1. Schematic of the USV–AUV acoustic cell. The USV is the unique sink. Inner AUVs (I1–I4) can reach it in one hop; outer AUVs (O1–O4) require a relay. Dashed purple segments are in-range acoustic hops; solid arrows are DATA unicasts; orange arrows are a HELLO broadcast. The drawing is a layout, not a measured snapshot.

First, inter-symbol interference. The symbol duration is 1 ms. A delay spread of tens of milliseconds, which Preisig and van Walree document as routine in littoral channels, spans many symbols. An energy SNR that ignores that spread overstates packet success.

Second, coherent fading. The same rays can cancel at the hydrophone even when the energy-based TL of (3) predicts a usable γ. A node that sits inside $R^\star$ can still fail to decode; a node that sits outside $R^\star$ can, rarely, succeed. Equation (4) is a contour, not a wall.

Bellhop ray tracing is the standard way to import that structure into a network simulator [27]. LQARP does so for *moored, static* nodes [17]. This paper does not. A mobile USV–AUV geometry would require a new ray trace at every HELLO, which we do not compute, and an offline TL table computed on a rest geometry is not a time-varying waveguide. The only multipath object in the model is the constant margin M in (3). It shortens $R^\star$ and stands in for unmodeled coherent loss. It is not a delay-spread process, not a tap filter, and not a substitute for sea-trial CSI. Section VI sweeps $M \in \{20, 30, 40\}$ dB as a sensitivity check: because nodes are placed as fractions of $R^\star$, discovery recall does not move in any operationally meaningful way. That is a statement about placement, not a claim that multipath is harmless.

D. Ambient Noise

Ambient noise in the band of interest is the four-component Wenz model, in dB re 1 μPa/Hz [4], [7], [17]:

$$N_{\mathrm{t}}(f) = 17 - 30\log_{10} f, \tag{5a}$$

$$N_{\mathrm{s}}(f) = 40 + 20(s - 0.5) + 26\log_{10} f - 60\log_{10}(f + 0.03), \tag{5b}$$

$$N_{\mathrm{w}}(f) = 50 + 7.5\sqrt{w} + 20\log_{10} f - 40\log_{10}(f + 0.4), \tag{5c}$$

$$N_{\mathrm{th}}(f) = -15 + 20\log_{10} f. \tag{5d}$$

The four terms are turbulence, shipping, wind-driven wave noise, and thermal noise. Shipping activity $s = 0.2$ and wind speed $w = 2$ m/s are taken from LQARP Table II. The noise level in a band of width B Hz is obtained by converting each component to linear units and summing,

$$\mathrm{NL} = 10\log_{10}\left(B \sum_{x\in\{\mathrm{t,s,w,th}\}} 10^{N_x(f)/10}\right). \tag{6}$$

Equation (6) is spatially homogeneous and temporally white in the packet band. There is no shipping-lane map, no rain term, no measured PSD, and no site-specific Wenz fit. It is the only noise the simulator uses, and it is the NL that enters (1).

E. Q-learning Technique

Q-learning is a model-free method for estimating the action-value function of a Markov decision process [11], [12]. An

agent in state s selects action a, receives a scalar reward r, and observes a successor s'. With learning rate $\alpha_Q \in (0, 1]$ and discount $\eta \in [0, 1)$—we write η rather than the usual γ, and α_Q rather than α, to avoid collision with SNR and with Thorp absorption—the Watkins update is

$$Q(s,a) \leftarrow (1-\alpha_Q)\, Q(s,a) + \alpha_Q \Big[r + \eta \max_{a'} Q(s',a') \Big]. \quad (7)$$

A greedy policy reads $\pi(s) = \arg\max_a Q(s,a)$. Fig. 2 is that loop, drawn as a routing agent.

In underwater next-hop routing the natural identification is [16], [17]

$$s = i, \qquad a = j \in \widehat{\mathcal{N}}_i(t), \qquad s' = j, \quad (8)$$

where i is the current holder of the packet and j is a neighbor the node has actually heard. The reward is protocol-specific. QELAR places residual energy in r [16]. LQARP adds an explicit link-quality term derived from the sonar equation, together with a depth term toward the sink [17]. A generic form that covers both, without inventing coefficients those papers use and we have not re-estimated, is

$$r(s,a) = \beta_{\mathrm{LQ}}\, r_{\mathrm{LQ}} + \beta_{\mathrm{z}}\, r_{\mathrm{z}} + \beta_{\mathrm{E}}\, r_{\mathrm{E}}, \quad (9)$$

with r_{LQ} increasing in SNR, r_{z} increasing as the next hop shallows toward the USV, and r_{E} increasing in residual energy. The weights $(\beta_{\mathrm{LQ}}, \beta_{\mathrm{z}}, \beta_{\mathrm{E}})$ belong to the protocol that trains (7). They are not fitted in this paper.

Two structural facts follow from (8) and are the reason this subsection exists.

First, the action set is the neighbor table. If j has never been heard, $Q(i,j)$ is not an entry that (7) is entitled to update. A learner trained on a geometric disc, or on an offline Bellhop table for moored nodes, is solving a different MDP than a mobile node whose table is an ALOHA process.

Second, greedy Q-learning, like greedy geographic routing, can stall in a *void*: a local minimum of the progress function from which every available next hop fails to approach the sink [14], [24], [25]. Write

$$\mathcal{P}_i(t) = \big\{ j \in \widehat{\mathcal{N}}_i(t) : \mathrm{prog}(j,0) > \mathrm{prog}(i,0) \big\} \quad (10)$$

for the progress set, with $\mathrm{prog}(\cdot, 0)$ a decreasing function of depth or of range to the USV. Node i is in a local void when $\mathcal{P}_i(t) = \emptyset$ while a path from i to the sink still exists in the true graph. Fig. 3 is that geometry. The greedy hop (dotted) points at a shallower node that is *not* a neighbor. The available neighbors (green) do not reduce depth or range. A detour through a non-greedy neighbor k can restore progress; LQARP's void handling is of that kind [17]. A network partition is a different object: if no path exists, no detour exists either.

This paper does not train (7) and does not run a void-recovery campaign. ETHA's next hop, defined in Section V, is a score on the live table, not a learned Q. The oracle bound in Section VI has zero no-route on the relay-stable field, which means that field is not a void field: the residual is missing table entries, not a local minimum of depth. Q-learning remains the intended sequel once that residual is the one that remains, and only among neighbors the node has actually heard.

III. Related Work

Section II defined the acoustic primitives. This section locates prior protocols relative to them.

A. Acoustic Channels and Link Abstraction

The sonar equation, Thorp absorption, practical spreading, and Wenz noise of Section II-B–II-D are standard [3]–[7]. Preisig and van Walree emphasize that real waveguides add coherent multipath and scattering that no k-spreading law captures [22], [23]. Bellhop ray tracing is the standard way to import that structure [27]. LQARP uses an offline Bellhop table for *static* moored nodes [17]. We do not. The constant margin M of Section II-C is the only multipath object in the simulator, and Section VI treats it as a sensitivity parameter, not as a ray trace.

B. Topology Discovery and Link-Quality Estimation

Terrestrial sensor nets learned that neighborhood is a measurement problem: EWMA of packet success, four-bit estimators, and hysteresis on SNR [28], [29]. Underwater, the same need is sharper because a HELLO occupies hundreds of milliseconds and a collision wastes a long propagation slot [19], [20], [26]. Periodic HELLO is the default in Aqua-Sim examples and in most routing evaluations [18]. Event-triggered and self-triggered control replace periodic sampling with a send-on-delta rule that guarantees a minimum inter-event time $T_{\min}$ and a keep-alive $T_{\max}$ [9], [10]. We import that structure to acoustic HELLOs and score it against ground-truth SNR-gated edges, not against PDR of a router that hides the table.

C. Geographic and Opportunistic Forwarding

VBF forwards inside a routing pipe toward the sink [13]; HH-VBF rebuilds the pipe per hop [15]; DBR and VAPR use depth or pressure and void awareness [14], [25]; GEDAR adds depth adjustment [24]. All of them need a current neighbor set. None of them is the experiment in this paper. We use geographic progress and a DBR-like depth score only so that discovery quality is visible in PDR, energy, stale hops, and no-route drops.

D. Learning Routers

QELAR treats forwarding as a distributed MDP with residual energy in the reward [16]. LQARP adds an explicit link-quality term derived from the sonar equation and evaluates Q-learning next-hop selection on a static UWSN with ALOHA and a Bellhop TL table [17]. Section II-E recorded that loop without training it. Those papers are the baseline we follow for *parameters* (Table II of LQARP) and for the intended venue. They are not the algorithm we run. Table I states the difference in one place. Importing Q-learning onto a map whose event-triggered recall is 0.84 would attribute discovery loss to the learner. Section VI makes that residual explicit.

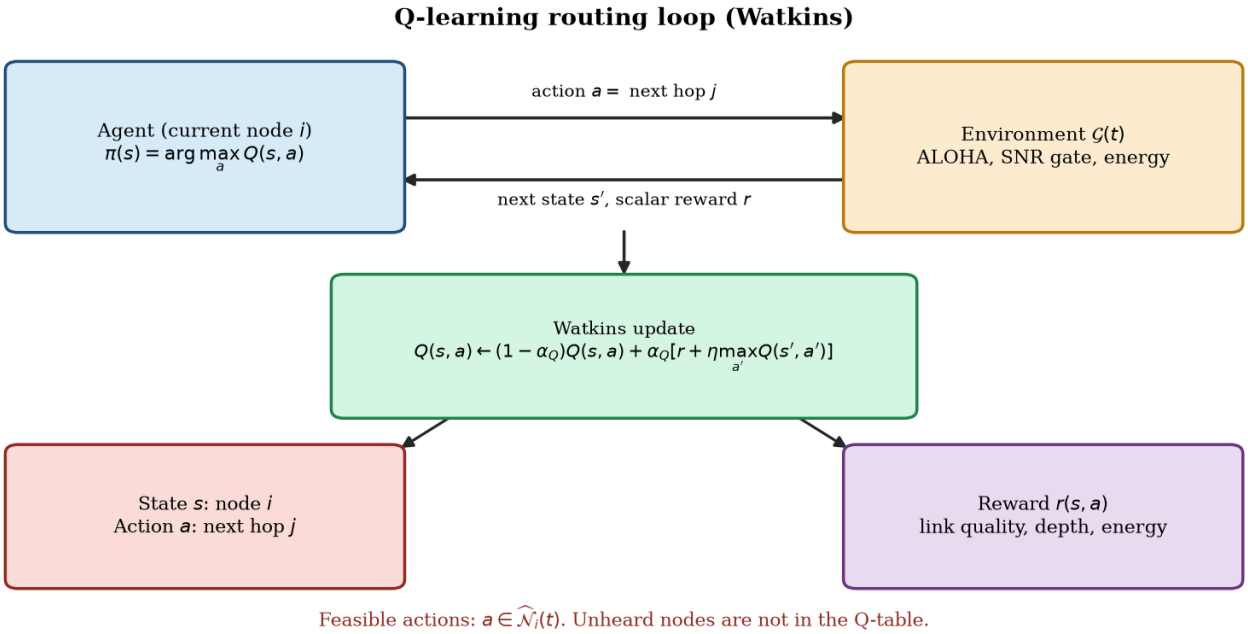


Fig. 2. Q-learning framework for next-hop routing (Watkins update (7)). The agent is the current holder of the packet; the environment is the acoustic network; the action is a next hop. Feasible actions are heard neighbors only. This figure is the loop trained by QELAR and LQARP. It is not a loop trained in this paper.

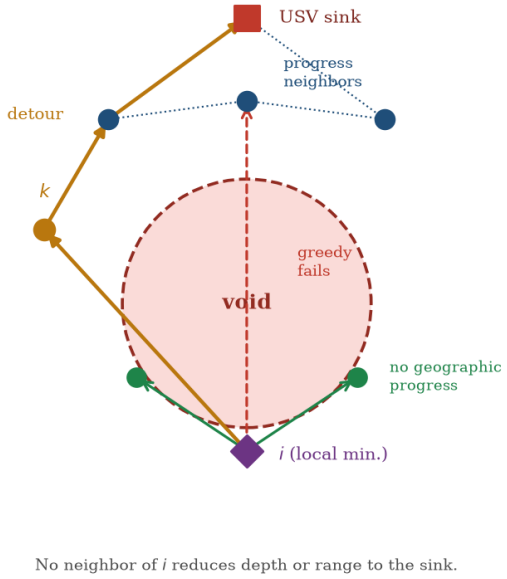


Fig. 3. Schematic of a geographic void. Node i is a local minimum: its heard neighbors do not reduce depth or range to the USV, so a greedy progress rule fails. A non-greedy neighbor k restores a path. The drawing is pedagogical. It is not a layout from the Aqua-Sim campaign, and no void-recovery PDR is reported.

TABLE I
POSITION RELATIVE TO CLOSELY RELATED PROTOCOLS

	DBR	QELAR	LQARP	ETHA
HELLO process scored	no	no	no	**yes**
Event-triggered beacons	no	no	no	**yes**
SNR-gated decode	range	energy MDP	sonar+Bellhop	sonar, no Bellhop
Hold-time feature	no	no	no	**yes**
Depth toward sink	yes	reward	reward	depth-primary
Q-learning	no	yes	yes	no (future)
Mobility in evaluation	limited	static energy	static UWSN	MCM+AUV thrust

IV. SYSTEM MODEL

This section instantiates Section II as the objects Aqua-Sim-NG integrates: the time-varying graph, the pairwise SNR, the BPSK erasure, Caruso kinematics, unslotted ALOHA, and transmit energy. Notation follows LQARP's acoustic stack [17]. The development remains analytical: there is no Bellhop field and no sea-trial CSI. Table II lists the principal symbols.

A. Network Model

Consider a set $\mathcal{V} = \{0, 1, \ldots, N\}$ of half-duplex acoustic nodes with the architecture of Fig. 1. Index 0 is the USV sink; indices $1, \ldots, N$ are AUVs. A directed pair $(i, j) \in \mathcal{V} \times \mathcal{V}$, $i \neq j$, is a *true link* at time t if and only if the pairwise SNR of Section II-B satisfies $\gamma_{ij}(t) \geq \gamma_{\text{th}}$. The true graph is

$$\mathcal{G}(t) = \big(\mathcal{V},\ \{(i,j) : i \neq j,\ \gamma_{ij}(t) \geq \gamma_{\text{th}}\}\big). \qquad (11)$$

Each node i maintains a neighbor table $\widehat{\mathcal{N}}_i(t)$ from HELLO receptions. Discovery quality is the agreement between $\widehat{\mathcal{N}}_i(t)$ and the true neighborhood of i in $\mathcal{G}(t)$. Data packets originated by AUVs have destination 0. Nodes share a single carrier and cannot transmit and receive at once.

Fig. 4 shows the inner/outer placement used for forwarding. Inner AUVs lie in $[0.62, 0.82]R^\star$ and remain usable relays.

TABLE II
PRINCIPAL NOTATION

Symbol	Meaning
$\mathcal{V}$, N	Node set; number of AUVs
$\mathcal{G}(t)$	True SNR-gated directed graph at time t
$\widehat{\mathcal{N}}_i(t)$	Neighbor table of node i
d_{ij}, p_i, v_i	Range (m); position; velocity
$\alpha(f)$, k, M	Thorp absorption; spreading; multipath margin
SL, TL, NL, DI	Sonar-equation terms (dB)
γ_{ij}, γ_{th}	Instantaneous SNR; decode threshold
$R^\star$	Range at which $\gamma = \gamma_{\text{th}}$
p_{e}, P_{succ}	BPSK BER; packet success probability
v_r, f_d	Radial relative speed; Doppler (Hz)
$T_{\min}$, $T_{\max}$	Event-triggered inter-HELLO bounds
$\bar{s}_{ij}$, H_{ij}	EWMA SNR; predicted hold (s)
O_c, E_{tx}	Control rate (bit/s); transmit energy (J)
$Q(s,a)$, α_Q, η	Action value; learning rate; discount
$\mathcal{P}_i(t)$	Progress-neighbor set of node i

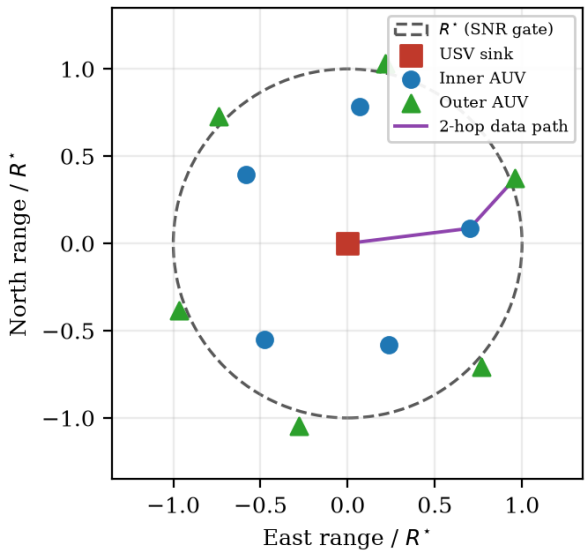


Fig. 4. Inner/outer forwarding layout. Ranges are normalized by the SNR-threshold range $R^\star$. Outer AUVs need a two-hop path through the inner ring to the USV sink.

Outer AUVs lie in $[1.02, 1.10]R^\star$ and require two hops. A separate discovery layout places both rings on the $R^\star$ contour so that links form and break within 240 s.

B. Acoustic Channel Instantiation

The simulator integrates (1)–(6) at $f_c = 24\,\text{kHz}$ with the Table II constants of Section II-B–II-D. Let $d_{ij}(t) = \|p_i(t) - p_j(t)\|_2$. The pairwise SNR is

$$\gamma_{ij}(t) = \text{SL} - \text{TL}\big(d_{ij}(t)\big) - \text{NL} + \text{DI}, \tag{12}$$

and a packet is *decodable* when $\gamma_{ij}(t) \geq \gamma_{\text{th}}$. The range derivative used by the hold-time predictor is

$$\frac{\partial \text{TL}}{\partial d} = \frac{10k}{d \ln 10} + \frac{\alpha(f)}{1000} \quad \text{(dB/m)}. \tag{13}$$

Aqua-Sim's geometric transmit range is set to $1.35R^\star$ so that the MAC can deliver a waveform that the SNR gate subsequently rejects; those events are logged, not counted as neighbors. Fig. 5 plots (2)–(15) at the operating point: absorption, Wenz components, TL for three values of M, and SNR together with P_{succ}. P_{succ} collapses in a few hundred metres around $R^\star$; that cliff is the decode gate used in every experiment.

C. Packet-Error Model

After ALOHA has delivered a waveform, BPSK bit errors follow LQARP eqs. (36)–(38) with constellation size $M = 2$ [17]:

$$p_{\text{e}}(\gamma) = \frac{1}{2}\,\text{erfc}\sqrt{10^{\gamma/10}}. \tag{14}$$

An L-bit packet succeeds with probability

$$P_{\text{succ}}(\gamma) = \big(1 - p_{\text{e}}(\gamma)\big)^L, \qquad L = 400, \tag{15}$$

and is erased independently with probability $1 - P_{\text{succ}}$. The same gate is applied to HELLO and to unicast data, so a next hop that is already below γ_{th} cannot contribute to PDR. Acoustic Doppler

$$f_d = f_c v_r / c, \qquad v_r = \frac{(p_j - p_i)^\top (v_j - v_i)}{d_{ij}}, \tag{16}$$

with $c = 1500\,\text{m/s}$, is logged from the radial relative speed and is not used as an additional decode threshold.

D. Mobility Model

Horizontal drift follows the meandering-current model (MCM) of Caruso *et al.* [8]. With wavelength λ, nondimensional coordinates $x' = x/\lambda$, $y' = y/\lambda$, meander amplitude $B'(t) = (B/\lambda)(1 + \varepsilon \cos \omega t)$, and $k_\psi = 2\pi$, the stream function is

$$\psi(x', y', t) = -\tanh\left(\frac{y' - B'(t)\sin(k_\psi x')}{\sqrt{1 + k_\psi^2 B'(t)^2 \cos^2(k_\psi x')}}\right). \tag{17}$$

The current is $u = U_0 \partial\psi/\partial y'$, $v = -U_0 \partial\psi/\partial x'$ with U_0 drawn per seed in $[0.30, 0.60]\,\text{m/s}$. Each AUV i adds a bounded thrust u_i^{th}, so

$$\dot{p}_i(t) = \big(u(p_i, t),\, v(p_i, t)\big) + u_i^{\text{th}}. \tag{18}$$

Depth is drawn once in $[40, 160]\,\text{m}$ for AUVs and held; the USV sits at $z = 8\,\text{m}$. This is kinematics, not an ocean general circulation model and not a vehicle autopilot.

E. MAC and Energy Consumption

Nodes share unslotted ALOHA with $\text{AckOn} = 0$ at bit rate $R_b = 1000\,\text{bit/s}$ [17], [31]. Packet duration is

$$T_{\text{tx}} = L/R_b = 0.4\,\text{s}. \tag{19}$$

Propagation at $R^\star$ is $T_{\text{prop}} = R^\star/c \approx 3.3\,\text{s}$, so a one-hop flight is about 3.7 s. Transmit energy for N_{tx} packets (HELLO plus data originations and forwards) at electrical power $P_T = 10\,\text{W}$ is

$$E_{\text{tx}} = P_T T_{\text{tx}} N_{\text{tx}} = 4N_{\text{tx}} \quad \text{(joules)}. \tag{20}$$

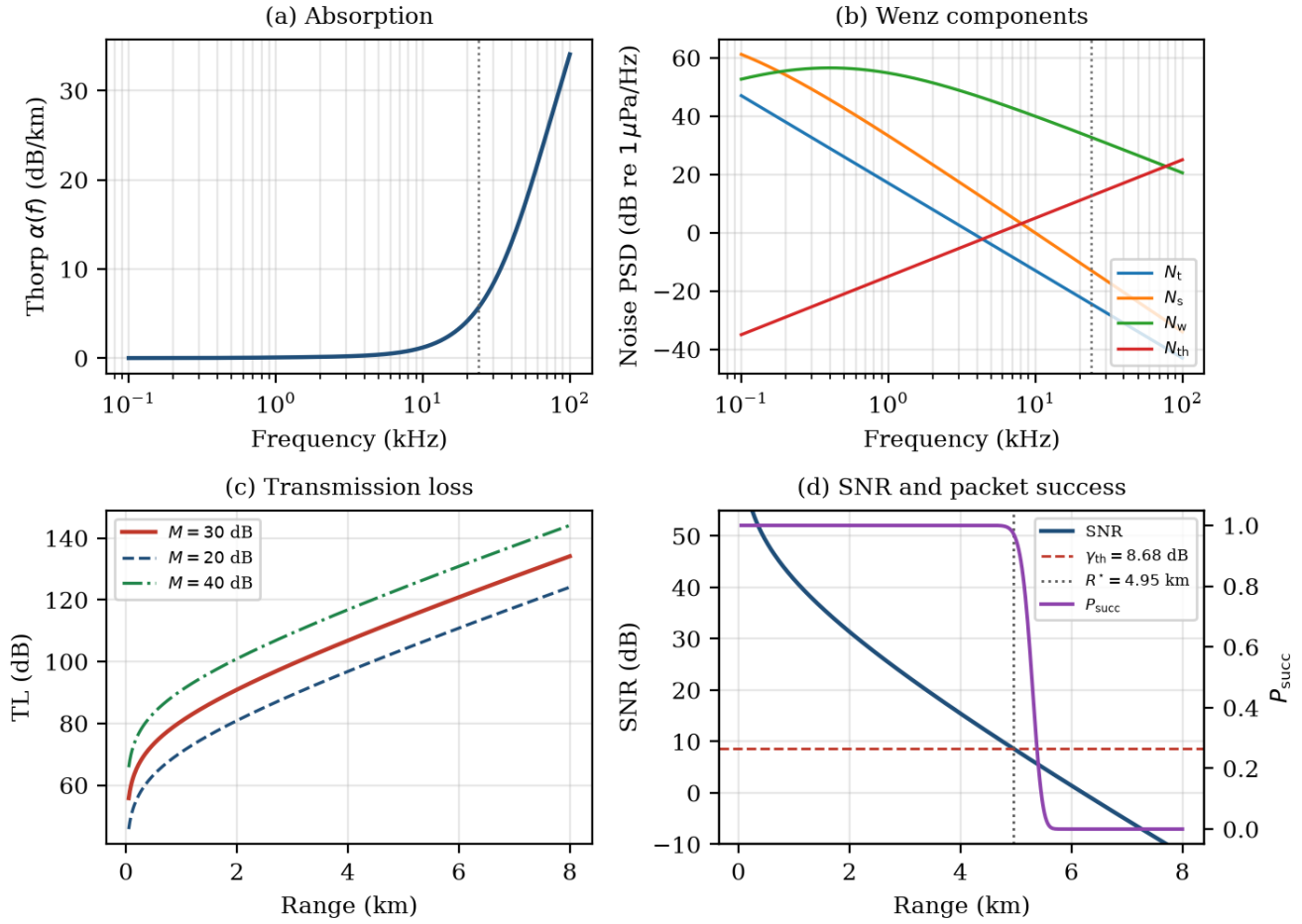


Fig. 5. Closed-form acoustic stack used by the simulator (LQARP Table II parameters, $f_c = 24\,\text{kHz}$, $k = 1.5$, $s = 0.2$, $w = 2\,\text{m/s}$, $M = 30\,\text{dB}$ unless swept). (a) Thorp absorption. (b) Wenz noise PSD components. (c) Transmission loss. (d) SNR and BPSK packet success versus range, with the SNR threshold γ_{th} and $R^\star$. This is not Bellhop.

Receive power is not an independent Aqua-Sim-NG parameter in this build; reported energy is therefore transmit energy only. Control rate against a 2 kbps reference budget is

$$O_c = \frac{N_{\text{HELLO}} L}{T_{\text{sim}}}. \tag{21}$$

A slotted-ALOHA control with slot $T_{\text{tx}} + T_{\text{prop,max}} + \text{guard} \approx 5.06\,\text{s}$ is reported as a negative result in discovery: the slot synchronizes the 5 s HELLO period and *raises* collisions.

F. Problem Statement

Over a horizon $[0, T_{\text{sim}}]$, each node must choose HELLO instants and, on each data packet, a next hop $j \in \widehat{\mathcal{N}}_i(t) \cup \{\emptyset\}$. Writing π_{HELLO} for the beacon policy and π_{hop} for the hop rule, the design problem is

$$\begin{aligned} \max_{\pi_{\text{HELLO}},\, \pi_{\text{hop}}} \quad & \mathbb{E}[\text{PDR}] \\ \text{s.t.} \quad & O_c \le 2\,\text{kbps}, \\ & j \in \widehat{\mathcal{N}}_i(t) \cup \{\emptyset\}, \end{aligned} \tag{22}$$

where the expectation is over seeds (placement, current, ALOHA, BPSK). We do not solve (22) as a global MDP. We specify an event-triggered HELLO rule and a hold-time-aware hop filter, then measure the resulting discovery and delivery residuals against periodic HELLO, SNR-greedy forwarding, a DBR-like depth rule, and an oracle with perfect geometry.

V. Proposed ETHA Protocol

ETHA (event-triggered hold-time-aware discovery and forwarding) has four parts: the neighbor record, the HELLO policy, the hold-time predictor, and the next-hop rule. Fig. 6 is the architecture; Fig. 4 is the topology on which the hop rule is evaluated; discovery is evaluated on the contour-crossing layout of Section VI.

A. Neighbor Record and EWMA Link Quality

When i decodes a HELLO from j, it stores last range, velocity, and SNR s_{ij}. Link quality is the EWMA [28]

$$\bar{s}_{ij}(t) = \lambda s_{ij}(t) + (1-\lambda)\bar{s}_{ij}(t^-), \qquad \lambda = 0.35, \tag{23}$$

with slope $\dot{\bar{s}}_{ij} = (\bar{s}_{ij}(t) - \bar{s}_{ij}(t^-))/\Delta t$. Neighbor j remains in $\widehat{\mathcal{N}}_i(t)$ while $t - t_{ij}^{\text{rx}}$ is within a timeout

$$\tau_{ij} = \min\{\tau_{\text{base}},\, \max(5, H_{ij} + 4),\, \tau_{\text{fade}}\}, \tag{24}$$

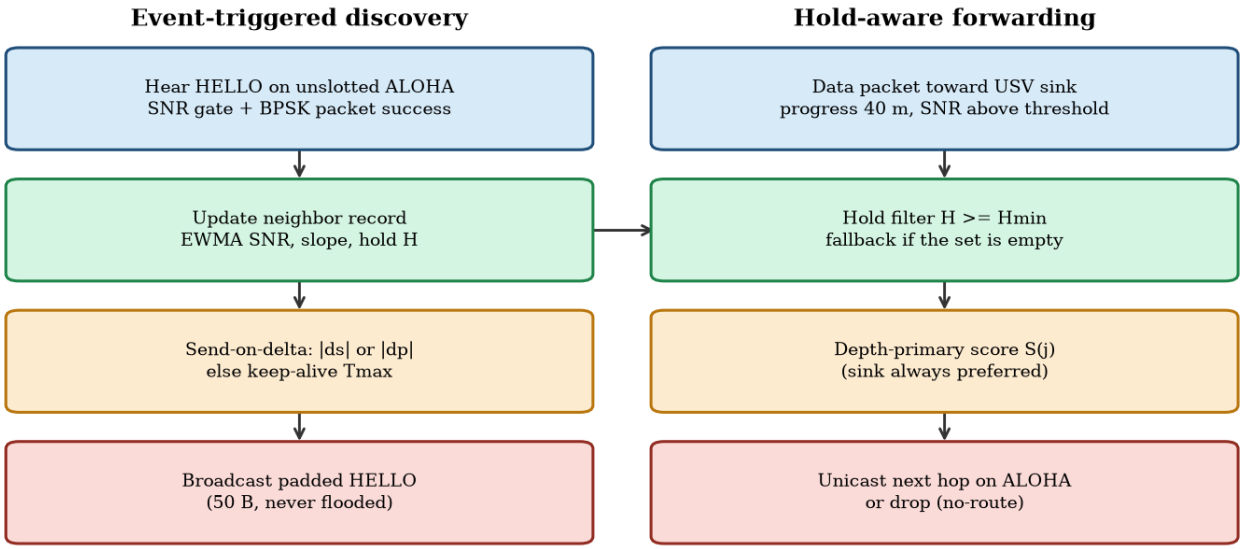


Fig. 6. ETHA architecture. Left: event-triggered, SNR-gated HELLO on unslotted ALOHA. Right: hold-time filter and depth-primary next hop toward the USV, using only neighbors the node has actually heard.

where $\tau_{\text{base}} = 3T$ (periodic) or $2T_{\max}$ (event), H_{ij} is the predicted hold, and $\tau_{\text{fade}} = 8\,\text{s}$ if $\bar{s}_{ij} < \gamma_{\text{th}} + 3\,\text{dB}$ with negative slope. Equation (24) is why link-down delay does not track $T_{\max}$.

B. Event-Triggered HELLO

Periodic mode emits a HELLO every $T = 5\,\text{s}$ plus a few hundred milliseconds of jitter.

Event-triggered mode is send-on-delta [9], [10]. Let t_ℓ be the last transmission, p_ℓ the position then, and s_ℓ the SNR to the USV then. A HELLO is eligible only if $t - t_\ell \geq T_{\min}$ ($T_{\min} = 5\,\text{s}$) and is forced if $t - t_\ell \geq T_{\max}$. Between those bounds i transmits when

$$\left|s_{\text{USV}}(t) - s_\ell\right| \geq \Delta_s \quad \text{or} \quad \|p_i(t) - p_\ell\| \geq \Delta_p, \tag{25}$$

with $\Delta_s = 4\,\text{dB}$ and $\Delta_p = 150\,\text{m}$ (Algorithm 1). Outer nodes whose USV SNR is already below γ_{th} do not treat a dead USV link as a permanent event; they fall back to $T_{\max}$ so the ring does not beacon every $T_{\min}$.

HELLOs carry identity, position, and velocity (28 bytes) and are padded to 50 bytes. They are sent through the Aqua-Sim device so ALOHA collisions are real. They are never flooded.

C. Hold-Time Predictor

Let v_r in (16) be positive when range opens. Geometric remaining time is

$$H_{\text{geom}} = \begin{cases} 0, & v_r > 0.02,\ d \geq R^\star, \\ (R^\star - d)/v_r, & v_r > 0.02,\ d < R^\star, \\ +\infty, & \text{otherwise.} \end{cases} \tag{26}$$

Algorithm 1 Event-triggered HELLO at node i

$t_\ell \leftarrow -\infty$, $p_\ell \leftarrow p_i$, $s_\ell \leftarrow -\infty$
loop
  wait until $t - t_\ell \geq T_{\min}$
  $s \leftarrow$ SNR to USV; $\Delta p \leftarrow \|p_i - p_\ell\|$
  if $t - t_\ell \geq T_{\max}$ **or** ($s \geq \gamma_{\text{th}}$ **and** ($|s - s_\ell| \geq \Delta_s$ **or** $\Delta p \geq \Delta_p$)) **then**
    broadcast padded HELLO on ALOHA
    $t_\ell \leftarrow t$, $p_\ell \leftarrow p_i$, $s_\ell \leftarrow s$
  end if
end loop

SNR remaining time uses the EWMA slope when the link is fading, and the kinematic slope $-\big(\partial\text{TL}/\partial d\big)v_r$ from (13) otherwise:

$$H_{\text{SNR}} = \begin{cases} 0, & \bar{s} < \gamma_{\text{th}}, \\ (\bar{s} - \gamma_{\text{th}})/(-\dot{\bar{s}}), & \dot{\bar{s}} < -10^{-3}, \\ (\bar{s} - \gamma_{\text{th}})/\big((\partial\text{TL}/\partial d)\, v_r\big), & v_r > 0.02, \\ +\infty, & \text{otherwise.} \end{cases} \tag{27}$$

The predictor outputs

$$H_{ij} = \min\big(H_{\text{geom}}, H_{\text{SNR}}, 300\big). \tag{28}$$

Fig. 7 plots $H_{\text{geom}} = (R^\star - d)/v_r$ against remaining range. At $H_{\min} = 8\,\text{s}$ the filter only binds in the last 2.4–12 m of opening range for $v_r \in [0.3, 1.5]\,\text{m/s}$, which is why it rarely moves PDR on the relay-stable forwarding field. This is first-order kinematics, not a learned remaining-useful-life model. A later Q-learner in the style of LQARP is entitled to use H_{ij} as a feature; it is not entitled to invent neighbors that were never heard.

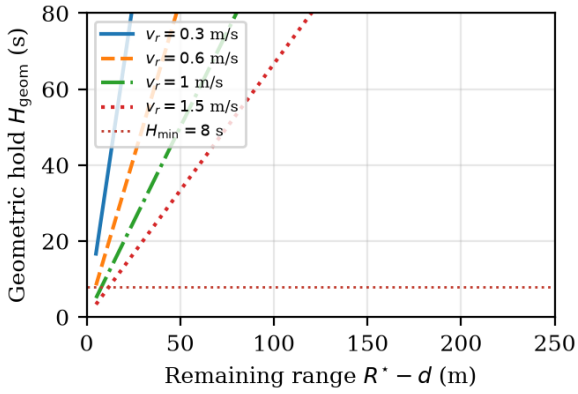


Fig. 7. Geometric hold time versus remaining range to $R^\star$. The $H_{\min} = 8\,\text{s}$ line is about two in-flight times at $R^\star$.

D. Hold-Time-Aware Next Hop

A next hop j must satisfy $\bar{s}_{ij} \geq \gamma_{\text{th}}$ (table) or $\gamma_{ij} \geq \gamma_{\text{th}}$ (oracle). Relays other than the sink must reduce range to the USV by at least $40\,\text{m}$, using the last HELLO position. ETHA then applies a hold-time constraint $H_{ij} \geq H_{\min} = 8\,\text{s}$ (about two in-flight times at $R^\star$) and a depth-primary score toward the surface sink

$$S(j) = \begin{cases} 10^6 + \bar{s}_{ij}, & j = 0, \\ -z_j + 0.02\,\bar{s}_{ij} + 0.01\min(H_{ij}, 120), & j \neq 0. \end{cases} \tag{29}$$

If the filtered set is empty, ETHA falls back to (29) without the hold test so a packet is not dropped solely because every neighbor is fading (Algorithm 2).

Four policies are compared.

- **ETHA (proposed).** Event-triggered HELLO, hold filter, score (29) on the live table.
- **Event/Periodic+SNR.** Maximize $\bar{s}_{ij}$ over the live table.
- **Event+Depth.** DBR-like [14]: maximize $-z_j$ among progress-feasible neighbors. This is an ablation of the hold term in (29).
- **Oracle.** ETHA score over *all* nodes using true geometry. The oracle cannot beat physics: a node with no currently decodable progress neighbor still drops.

E. Complexity

Let $n_i = |\widehat{\mathcal{N}}_i|$. A HELLO decision is $O(1)$: compare SNR and displacement against Δ_s and Δ_p. Updating one neighbor record is $O(1)$ (EWMA plus (28)). A forwarding decision scans the live table once, hence $O(n_i)$, and $n_i \leq N$ with $N \leq 15$ in the campaign. There is no Q-table, no value iteration, and no end-to-end path computation. Memory is one record per heard neighbor.

VI. Performance Evaluation

A. Simulation Settings

All experiments use NS-3.41 with Aqua-Sim-NG (commit `0fbc166`) [30], [31]. Table III lists the acoustic stack (LQARP Table II [17]). Table IV lists the two campaigns.

Algorithm 2 Table-driven next hop toward sink d

$\mathcal{N} \leftarrow$ live neighbors of i
$j^\star \leftarrow \emptyset$, best $\leftarrow -\infty$
for all $j \in \mathcal{N}$ **do**
  require $\bar{s}_{ij} \geq \gamma_{\text{th}}$
  require $j = d$ or geographic progress $\geq 40\,\text{m}$
  if ETHA **and** $H_{ij} < H_{\min}$ **then**
    continue
  end if
  score $\leftarrow S(j)$ in (29) (ETHA) or $\bar{s}_{ij}$ (SNR) or $-z_j$ (depth)
  if score $>$ best **then**
    $j^\star \leftarrow j$, best $\leftarrow$ score
  end if
end for
if $j^\star = \emptyset$ **and** ETHA **then** repeat without the hold test
return $j^\star$ or drop (no-route)

TABLE III
Acoustic and MAC Parameters (LQARP Table II Stack)

Parameter	Value
Carrier f_c / bit rate R_b	24 kHz / 1000 bit/s
Packet length L	400 bit
Modulation	BPSK
SL / DI	185 dB / 0 dB
Spreading k	1.5
Shipping s / wind w	0.2 / 2 m/s
SNR threshold γ_{th}	8.68 dB
Multipath margin M	30 dB ($R^\star = 4946\,\text{m}$)
Sound speed c	1500 m/s
Transmit power P_T	10 W
MAC	unslotted ALOHA, AckOn = 0
HELLO / data pad	50 B
Control budget (reference)	2 kbps

TABLE IV
Simulation Campaigns

Quantity	Discovery	Forwarding (ETHA)
Simulator	NS-3.41 + Aqua-Sim-NG	
Nodes	12 (contour)	6, 12, 16
Run length	240 s	360 s
Periodic HELLO	$T = 5\,\text{s}$	$T = 5\,\text{s}$
Event HELLO	$T_{\max} = 25\,\text{s}$	$T_{\max} = 10\,\text{s}$
Data	—	every AUV, 8–24 s, $t \geq 50\,\text{s}$
Seeds (main)	30	30
Seeds (scale / load / appendix)	20	20 / 15

Each seed randomizes placement, thrust, current phase, current scale, ALOHA backoff, and BPSK erasures. Intervals are two-sided 95% Student-t.

Discovery uses a contour-crossing layout so that true links form and break inside 240 s. Forwarding uses a relay-stable inner ring so that PDR measures the hop rule rather than a forced partition; data start at $t = 50\,\text{s}$ after neighbor tables have filled.

B. Performance Metrics

Let $\mathcal{E}(t)$ and $\widehat{\mathcal{E}}(t)$ be the true and discovered directed-edge sets at snapshot t. With $\text{TP} = |\mathcal{E} \cap \widehat{\mathcal{E}}|$, $\text{FP} = |\widehat{\mathcal{E}} \setminus \mathcal{E}|$, $\text{FN} =$

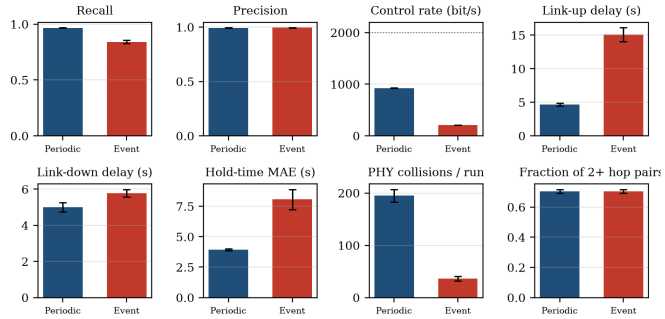


Fig. 8. Discovery metrics, mean and 95% CI over 30 seeds. Event-triggered HELLO stays near 11% of a 2 kbps budget. Precision is unchanged; recall, link-up delay, and hold-time MAE pay for the saved beacons. Link-down delay does not, because the timeout is SNR-aware.

TABLE V
DISCOVERY, 30 SEEDS, MEAN ± 95% CI

Metric	Periodic	Event
Recall	0.966 ± 0.002	0.841 ± 0.015
Precision	0.994 ± 0.001	0.994 ± 0.001
F1	0.979 ± 0.001	0.908 ± 0.009
Control (bit/s)	927 ± 1	212 ± 1
Link-up delay (s)	4.67 ± 0.20	15.1 ± 1.0
Link-down delay (s)	5.00 ± 0.26	5.78 ± 0.22
Hold-time MAE (s)	3.93 ± 0.07	8.06 ± 0.83
PHY collisions / run	195 ± 12	36 ± 4
2+ hop pair fraction	0.705 ± 0.012	0.705 ± 0.012
Beacons sent / run	556 ± 1	127 ± 0.5

$|\mathcal{E} \setminus \widehat{\mathcal{E}}|$,

$$\mathrm{Prc} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad \mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_1 = \frac{2\,\mathrm{Prc}\,\mathrm{Rec}}{\mathrm{Prc}+\mathrm{Rec}}. \tag{30}$$

Link-up (link-down) delay is the time from a true edge appearing (disappearing) until the table agrees. Hold-time MAE is the absolute error of H_{ij} in the 60 s before a true link-down. Delivery metrics are

$$\mathrm{PDR} = \frac{N_{\mathrm{deliv}}}{N_{\mathrm{orig}}}, \qquad \overline{D} = \frac{1}{N_{\mathrm{deliv}}} \sum (t_{\mathrm{rx}} - t_{\mathrm{tx}}). \tag{31}$$

Energy is (20) counted over HELLO plus data transmissions. A hop is *stale* if true SNR is already below γ_{th} or true hold is shorter than flight time $d/c + T_{\mathrm{tx}}$, and *dying* if true hold is below $H_{\min}$.

C. Topology Discovery: Periodic versus Event-Triggered

Fig. 8 and Table V summarize the thirty-seed contour-crossing experiment ($T_{\max} = 25$ s).

Event-triggered discovery saves about 77% of HELLO transmissions ($O_c = 212$ versus 927 bit/s) and stays at 11% of the 2 kbps budget. Precision does not move: false neighbors are rare because decode is SNR-gated after the MAC. The costs are real. Recall drops by 0.125. Mean link-up delay grows from 5 s to 15 s, consistent with a 25 s keep-alive plus ALOHA loss. Hold-time MAE roughly doubles because the predictor is fed less often. Link-*down* delay does not inflate, which is (24): a fading EWMA is declared dead in about 6 s rather than $3T_{\max} = 75$ s. Every seed produced 5–14 true link-down events; the delays are not a two-event anecdote.

PHY collisions fall from 195 to 36 per run. That is a second budget: fewer HELLOs leave airtime for data. It is also why a learning router that ignores the HELLO process will misattribute collisions.

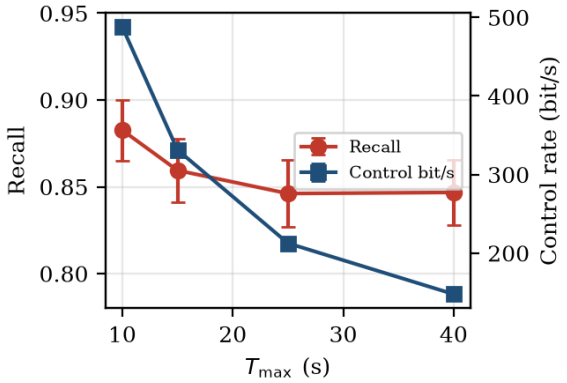


Fig. 9. Event-triggered keep-alive sweep (unslotted ALOHA, 12 nodes, $n = 20$). Recall saturates; control rate and hold-time error do not.

TABLE VI
EVENT-TRIGGERED $T_{\max}$ SWEEP ($n = 20$)

$T_{\max}$ (s)	Recall	Control (bit/s)	Hold MAE (s)
10	0.883 ± 0.018	488 ± 0.3	4.85 ± 0.13
15	0.859 ± 0.018	331 ± 0.7	5.55 ± 0.28
25	0.846 ± 0.019	213 ± 1	7.90 ± 0.98
40	0.847 ± 0.019	148 ± 1	13.9 ± 1.9

D. Impact of Keep-Alive, Slotting, Margin, and Scale

Fig. 9 and Table VI sweep $T_{\max}$ at $n = 20$ seeds. Stretching the keep-alive past 25 s does not buy recall; it only stale-dates hold time (MAE 13.9 s at $T_{\max} = 40$ s). $T_{\max} = 10$ s recovers recall 0.883 and hold MAE 4.85 s at 488 bit/s—still a quarter of a 2 kbps budget, and the awareness-friendly operating point used for forwarding. $T_{\max} = 25$ s is the budget-friendly discovery point.

Slotted ALOHA is worse in this geometry, not better. A UWA slot of 5.06 s lines up with the 5 s HELLO period. Periodic slotted recall falls to 0.818 and collisions rise to 477 per run. We keep unslotted ALOHA as the main MAC and report slotted as a negative control.

Multipath margin $M \in \{20, 30, 40\}$ dB rescales $R^\star \in \{6407, 4946, 3564\}$ m. Because nodes are placed as fractions of $R^\star$, recall, edge count, and the two-hop fraction do not change in any operationally meaningful way. This is a sensitivity check, not a substitute for Bellhop.

Node count matters for discovery. A 6-node ring is a near-clique (2+ hop fraction 0.26, diameter 1.47). Event-triggered recall collapses to 0.59 because everyone sits on the $R^\star$ contour and send-on-delta rarely fires. Twelve and sixteen nodes both have a 2+ hop fraction near 0.70 and event recall 0.85–0.86. We quote 12 nodes as the operating point and do not quote the six-node clique as evidence of a protocol.

TABLE VII
FORWARDING, 12 NODES, MEAN $\pm$ 95% CI ($n = 30$)

Scheme	PDR	E_{tx} (kJ)	O_c	Delay (s)	No-route
ETHA (proposed)	0.755 ± 0.031	2.82 ± 0.02	480	3.54 ± 0.07	12.7 ± 6.1
Event+SNR	0.651 ± 0.033	3.00 ± 0.03	480	3.74 ± 0.14	14.5 ± 7.0
Event+Depth	0.755 ± 0.031	2.82 ± 0.02	480	3.54 ± 0.07	12.7 ± 6.1
Periodic+SNR	0.641 ± 0.027	4.61 ± 0.03	927	3.70 ± 0.13	2.07 ± 1.2
Periodic+ETHA	0.739 ± 0.023	4.41 ± 0.02	926	3.51 ± 0.07	1.53 ± 0.71
Oracle	0.817 ± 0.027	2.81 ± 0.02	480	3.50 ± 0.07	0

E. Packet Delivery Ratio

All AUVs originate 400-bit packets every 16 s after $t = 50$ s on the relay-stable inner/outer field, $T_{\max} = 10$ s, 360 s, $n = 30$. Fig. 10 and Table VII are the main comparison.

ETHA delivers PDR $= 0.755$, a 18% relative gain over periodic SNR-greedy (0.641) and a 16% gain over event SNR-greedy (0.651) on the same event map. Mean hop count falls from 1.66 (SNR-greedy) to 1.37 (ETHA): depth-primary scoring walks toward the surface rather than toward a deep, high-SNR neighbor that is a geographic dead end. Stale in-flight hops are zero.

Event+Depth matches ETHA on every delivery statistic to the reported precision. That is a measured fact, not a hidden claim. On this relay-stable field the hold filter rarely binds ($H_{\min} = 8$ s, inner AUVs well inside $R^\star$), so (29) reduces to a DBR-like rule plus a vanishing SNR/hold tie-break. ETHA is not proposed as a replacement for depth when the sink is the surface; it is proposed as the way to *afford* that hop rule, by not spending 927 bit/s on HELLOs, and as the filter that would bind if the contour were crossed in flight.

The oracle with perfect geometry reaches 0.817 at the same event HELLO rate, with zero no-route. ETHA's remaining 0.062 of PDR is the 12.7 no-route drops per run from table holes, not hop scoring. Periodic HELLO almost eliminates no-route (2.07) but does not translate that map density into PDR under SNR-greedy scoring, because the extra 447 bit/s of HELLO also raise PHY collisions from 336 to 629 per run. Periodic+ETHA (0.739) is close to event ETHA (0.755) at 56% more transmit energy: once the scorer is depth-primary, further HELLOs buy little delivery and cost a great deal of airtime.

F. Energy Consumption

Transmit energy follows (20). ETHA spends 2.82 kJ per 360 s run versus 4.61 kJ for periodic SNR-greedy: a 39% saving. Table VIII splits that total into HELLO and data. Event schemes emit 432 HELLOs (480 bit/s), hence 1.73 kJ of beacon energy; periodic schemes emit about 834 HELLOs, hence 3.34 kJ. The remaining 1.09 kJ (ETHA) versus 1.28 kJ (periodic SNR-greedy) is data origination and forwarding. ETHA's shorter paths therefore save a further 0.19 kJ on top of the 1.61 kJ HELLO saving. Event+SNR sits at 3.00 kJ total; the extra 0.18 kJ relative to ETHA is additional data forwards on longer SNR-greedy paths, not extra HELLOs. Periodic+ETHA (4.41 kJ) shows that rewriting the hop score without rewriting HELLO still leaves about 1.6 kJ on the table. Fig. 11 (right) confirms the gap at every node count: at 16 nodes ETHA uses 3.93 kJ against 6.31 kJ for periodic SNR-greedy.

TABLE VIII
TRANSMIT-ENERGY DECOMPOSITION, 12 NODES, 360 S

Scheme	HELLO (kJ)	Data (kJ)	Total (kJ)
ETHA	1.73	1.09	2.82
Event+SNR	1.73	1.27	3.00
Periodic+SNR	3.34	1.28	4.61
Periodic+ETHA	3.34	1.07	4.41

TABLE IX
PDR AND ENERGY VERSUS NODE COUNT, MEAN $\pm$ 95% CI

	ETHA		Event+Depth		Periodic+SNR	
Nodes	PDR	E (kJ)	PDR	E (kJ)	PDR	E (kJ)
6 ($n = 20$)	0.875	1.25	0.875	1.25	0.865	2.12
12 ($n = 30$)	0.755	2.82	0.755	2.82	0.641	4.61
16 ($n = 20$)	0.657	3.93	0.657	3.93	0.525	6.31

G. End-to-End Delay and Hop Count

Mean e2e delay is one acoustic flight plus a short contention: 3.54 s for ETHA versus 3.74 s for event SNR-greedy and 3.70 s for periodic SNR-greedy. The differences are tens of milliseconds of extra hops and ALOHA backoff, not a routing-layer queue. Hop count 1.37 versus 1.66 is the geographic content of (29): more packets reach the USV in one hop because the scorer prefers the surface. Fig. 11 (bottom) repeats delay and hop count against node count: both rise with load, and ETHA stays below periodic SNR-greedy at every scale because it takes shorter surface-seeking paths.

H. Control Overhead

HELLO rate is deterministic in this campaign once $T_{\max}$ and T are fixed: 480 bit/s for every event scheme ($T_{\max} = 10$ s) and 927 bit/s for every periodic scheme ($T = 5$ s), against a 2 kbps reference. ETHA therefore occupies 24% of the control budget where periodic occupies 46%. Combined with Section VI-C, the operator has two documented points: $T_{\max} = 25$ s at 11% of budget for awareness, and $T_{\max} = 10$ s at 24% of budget for forwarding.

I. Scalability

Fig. 11 and Table IX scale the network at the same per-AUV load (one 400-bit packet every 16 s). PDR falls with node count for every scheme, as expected under unslotted ALOHA [19]. ETHA stays above periodic SNR-greedy at 6, 12, and 16 nodes (0.875, 0.755, 0.657 versus 0.865, 0.641, 0.525). The gap widens as collisions grow: at 16 nodes ETHA's HELLO rate is 640 bit/s versus 1236 bit/s periodic, and PDR is 0.13 higher. Six-node graphs are a dense reference; they are not the operating point we quote in the abstract.

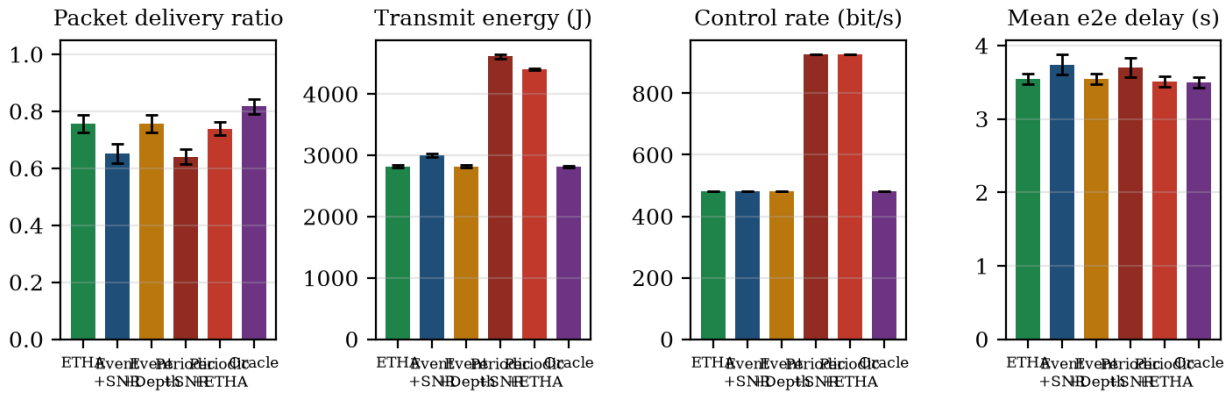


Fig. 10. Forwarding campaign, 12 nodes, $n = 30$, 95% CI. ETHA matches a DBR-like depth baseline on PDR and delay while cutting transmit energy and HELLO rate relative to periodic SNR-greedy forwarding. Oracle is an upper bound with perfect geometry, not a protocol.

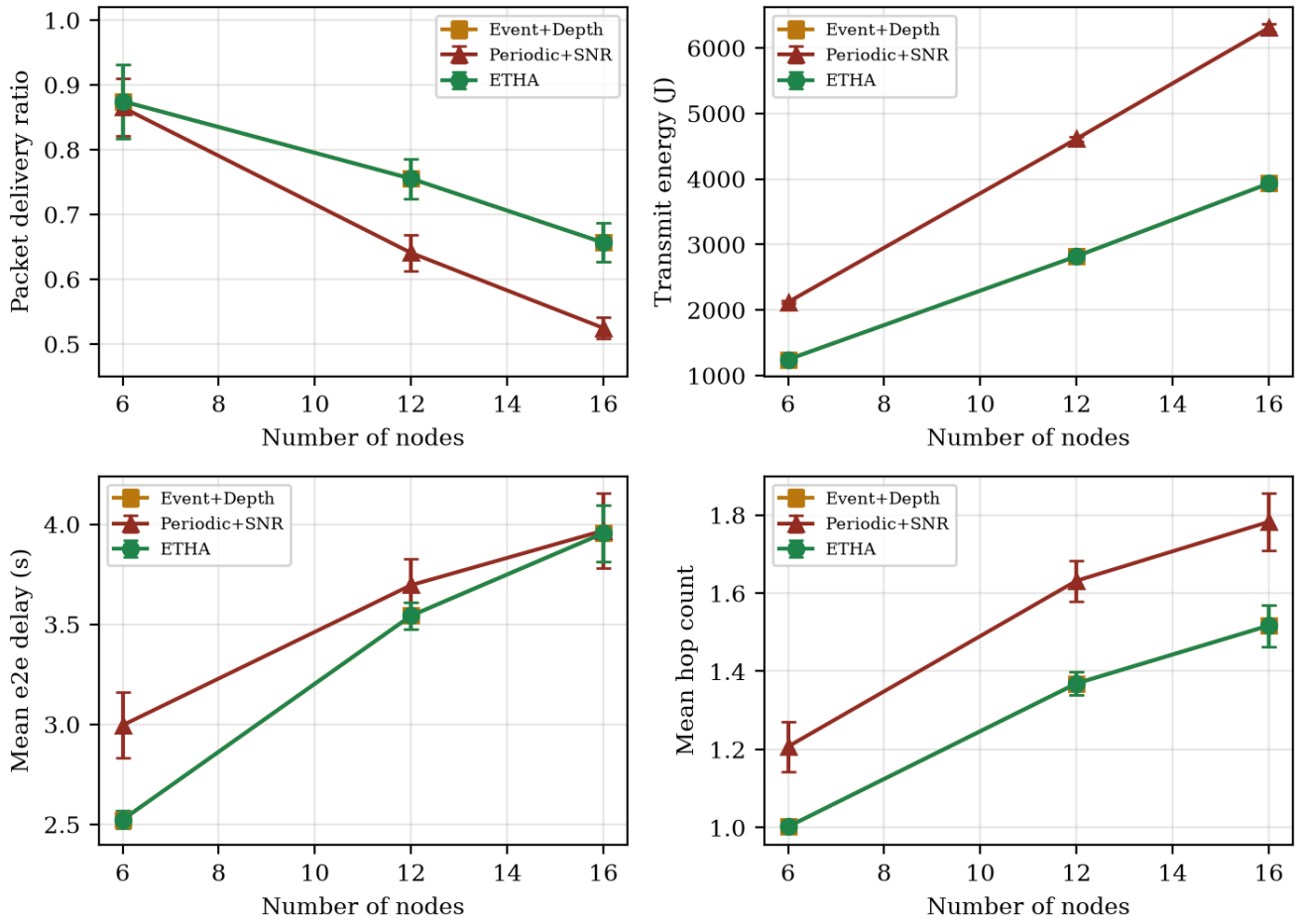


Fig. 11. LQARP-style scale plots (95% CI): packet delivery ratio, transmit energy, end-to-end delay, and hop count versus number of nodes. Load is one 400-bit packet per AUV per 16 s. ETHA and Event+Depth coincide on this field (hold rarely binds).

J. Impact of Traffic Load

Fig. 12 and Table X vary the packet generation interval in $\{8, 12, 16, 24\}$ s at 12 nodes ($n = 15$ except $n = 30$ at 16 s). ETHA PDR is essentially flat (0.746–0.755): at these offered loads the pipe is not saturated, and the first-order residual remains next-hop geography plus table holes. Periodic SNR-greedy stays 0.12–0.14 below ETHA at every interval and spends 1.7–2.0 kJ more. Tightening the interval from 24 s to 8 s raises ETHA energy from 2.47 kJ to 3.88 kJ because more data packets are originated and forwarded; the HELLO component does not change. Fig. 12 also reports delay and control rate versus interval: delay is insensitive to load in this

TABLE X
PDR VERSUS PACKET INTERVAL, 12 NODES

Interval (s)	ETHA	Event+Depth	Periodic+SNR
8 ($n = 15$)	0.746 ± 0.037	0.746 ± 0.037	0.611 ± 0.040
12 ($n = 15$)	0.755 ± 0.035	0.755 ± 0.035	0.620 ± 0.050
16 ($n = 30$)	0.755 ± 0.031	0.755 ± 0.031	0.641 ± 0.027
24 ($n = 15$)	0.746 ± 0.044	0.746 ± 0.044	0.625 ± 0.052

regime, and HELLO overhead does not grow with data rate because beacons are event-triggered, not piggybacked.

K. Discussion and Limitations

The first-order residual on a mobile USV–AUV acoustic pipe is the HELLO budget together with next-hop geography, not a learned Q-table. Event-triggered discovery at $T_{\max} = 10\,\mathrm{s}$ buys a neighbor table that is good enough for $\mathrm{PDR} = 0.755$ at 12 nodes while leaving more than $1.6\,\mathrm{kJ}$ and $447\,\mathrm{bit/s}$ on the table relative to periodic SNR-greedy HELLO. Hold-time-aware scoring is the correct filter when links die in flight; on the relay-stable forwarding field those events are rare, ETHA coincides with Event+Depth, and the paper reports that coincidence. Depth-greedy remains available as a hop rule on top of ETHA's map. Q-learning in the style of LQARP [16], [17] is the natural sequel once the map—not the scorer—is the residual that remains, and only among neighbors the node has actually heard.

The link model is analytical. There is no Bellhop TL table, no measured CSI, and no sea trial. Hold time is kinematic. Forwarding is single-path. Receive energy is not counted. The layout is engineered so that $R^\star$ is meaningful; a field whose nodes never approach the contour would show fewer link-down events and a weaker hold-time story. Those limits are why the claims stop at discovery plus a transparent hop filter.

VII. CONCLUSION

This paper wrote the missing first chapter of a mobile USV–AUV acoustic network that must live inside $2\,\mathrm{kbps}$ of control traffic. Section II defined the architecture, the Urick–Thorp–Wenz channel, the constant-margin treatment of multipath, and the Q-learning loop that LQARP trains—without training it. The system model instantiates that channel as (2)–(20) in Aqua-Sim-NG. Event-triggered HELLO at $T_{\max} = 25\,\mathrm{s}$ cuts discovery overhead from 927 to $212\,\mathrm{bit/s}$ at a measured cost in recall. The forwarding operating point $T_{\max} = 10\,\mathrm{s}$, together with hold-time-aware depth-primary next-hop selection (ETHA), raises PDR from 0.641 (periodic SNR-greedy) to 0.755 at 12 nodes ($n = 30$) while cutting transmit energy from $4.61\,\mathrm{kJ}$ to $2.82\,\mathrm{kJ}$ and HELLO traffic from 927 to $480\,\mathrm{bit/s}$. A DBR-like depth ablation on the same map matches ETHA and is reported as a baseline, not hidden. An oracle bound of 0.817 locates the remaining gap in missing relays. The next chapter is a learner that is only allowed to choose among neighbors ETHA has actually heard.

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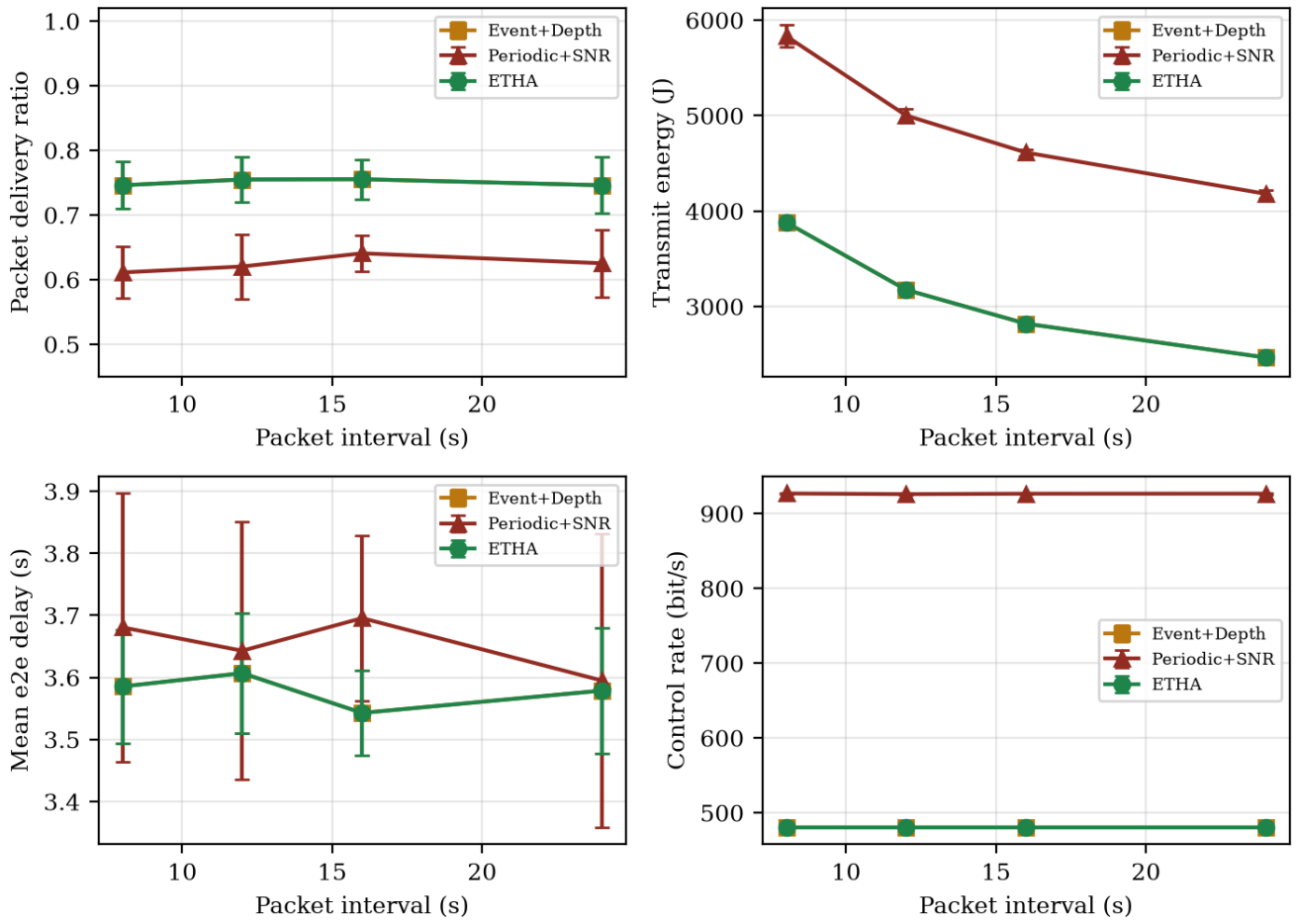


Fig. 12. LQARP-style load plots (95% CI): PDR, transmit energy, end-to-end delay, and HELLO control rate versus packet generation interval at 12 nodes.

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